

Problem Statement No. 24: Smart Satellite Communication Knowledge Assistant

THE CHALLENGE

Satellite communication systems involve complex concepts such as frequency bands, signal modulation, link budgets, and communication protocols. Engineers and students often struggle to understand these topics because the knowledge is distributed across textbooks, research papers, and technical documentation.

Finding relevant information and troubleshooting communication issues can take significant time.

The challenge is to build an intelligent assistant that can retrieve knowledge and explain satellite communication concepts in an easy and interactive manner.

THE OBJECTIVE

Build a Satellite Communication Knowledge Assistant that helps users understand and troubleshoot satellite communication systems.

The solution should:

1. Knowledge Retrieval

Extract information from satellite communication documents and technical materials.

2. Concept Explanation

Explain complex topics such as modulation techniques, frequency bands, and link budgets in simple language.

3. Troubleshooting Support

Provide guidance for solving satellite communication problems.

4. Conversational Interface

Allow users to ask questions and receive responses through an interactive chat interface.

TECH STACK

- LangFlow + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models

- Replicate with LangChain + RAG + IBM Granite Models